Band-1 Hadamard Order 6836 — Miyamoto (Paley-II Seed H_1708 q=853 + C_1710); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_1708 Paley-II PASS; C_1710 PASS; H_6836 Miyamoto PASS (NOT in SPK); gap 2052 still OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. 2026-09-15_Band1_Hadamard_6788_Miyamoto; 2026-09-15_Band1_Hadamard_6724_Miyamoto.
Lab. `/workspace/experiments/band1/hadamard_6836_miyamoto/`.
Cutoff. 2026-09-16T02:51:58Z (~10:51PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1708 and 6836 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-II seed H_{1708} = Paley-II($q = 853$) and symmetric Paley conference C_{1710} over prime $q = 1709 \equiv 1 \pmod 4$. Classical SPK for order **6836**: Paley-I **FAIL**s ($q = 6835 = 5 \cdot 1367$); Paley-II **FAIL**s ($q = 3417 = 3 \cdot 17 \cdot 67$); Kronecker **FAIL**s ($4 \times 1709$). Therefore order **6836 is a Miyamoto-only bonus (NOT in SPK)**. Seed **1708** enters SPK via classical Paley-II. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior skip $q = 1637 \to n = 6548$ retained; intervening after $q = 1697$: $q = 1701/1705$ not prime; $q = 1709$ next buildable. Prior SPK seeds **840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/6788/6836/...** excluded.

Abstract

Finite constructions of orders **1708** and **6836**.

- H_{1708} = Paley-II ($q = 853$)
- C_{1710} = Paley conference over prime $q = 1709$
- H_{6836} = Miyamoto(H_{1708}, C_{1710})
- Paley-I / Paley-II / Kronecker for 6836: **SKIP / FAIL**

SPK: 6836 Miyamoto-only bonus (NOT in SPK). Seed 1708 enters SPK. Prior SPK **840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained. **Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6836.py`

Results. overall_pass=true; units 93/93; wall 256.8435s; all max|HHˆT-nI|=0.

Cutoff: 2026-09-16T02:51:58Z (~10:51PM ET)

Arithmetic

- Target $n = 6836 = 4 \cdot 1709$; 1709 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1708} and C_{1710} (Paley conference over prime $q = 1709$).
- Seed: Paley-II $q = 853$ (prime $\equiv 1 \pmod 4$) → $H_{1708} = 2(q + 1)$.
- Seed Paley-I $q = 1707 = 3 \cdot 569$ **FAIL**; Double half=854 **FAIL** ($q = 853 \equiv 1 \pmod 4$).
- SPK: Paley-I $q = 6835$ **FAIL**; Paley-II $q = 3417$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only (NOT in SPK)**.
- Prior skip: $q = 1637 \rightarrow n = 6548$. Intervening: $q = 1701/1705$ not prime; $q = 1709$ next buildable after 1697.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1708 Paley-II	Paley-II q=853	0	2.6372
C_1710	Paley conf q=1709	0	3.4452
H_6836 Miyamoto	Miyamoto(H_1708 Paley-II, C_1710)	0	247.7332
H_6836 Paley-I	SKIP q=6835=5 · 1367 not prime	—	—
H_6836 Paley-II	SKIP q=3417=3 · 17 · 67 not prime	—	—
H_6836 Kronecker	SKIP 4 · 1709 not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **6788** lineage (**6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 6676/6484/6404/6388/6196/6788/6836) + verified **1708** (Paley-II classical); **exclude** Miyamoto-only bonuses including **6836**. Known count 1314. Next constructive gap **2052** (OPEN).

Quantity	Value
6836 in SPK	**False** (Miyamoto-only)
1708 in SPK	**True** (Paley-II q=853)
840 / 1680 / 6724 in SPK	**True** (prior)
212 / 424 / 1696 in SPK	**True** (prior)
1668 / 6628 / 1656 / 1620 / 1608 / 6436 in SPK	**True** (prior)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · 6836 is NOT folded into SPK.